

External quantities involving lattice displacement β

$$A_{\mu, st}(\mathbf{k}) = \langle u_s^W | i \partial_\mu | u_t^W \rangle \quad [\text{AA}], \quad (16)$$

D.V. 6/23/26

$$A_{\mu, st}(\mathbf{k}) = \sum_{\mathbf{R}} e^{i\mathbf{k} \cdot \mathbf{b}_{st}(\mathbf{R})} X_{\mu, st}(\mathbf{R}), \quad [\text{ft-aa}] \quad (18)$$

$$X_{\mu, st}(\mathbf{R}) = \langle 0s | \hat{r}_\mu - \tau_{s\mu} | \mathbf{R}t \rangle, \quad [\text{r-real}] \quad \leftarrow \text{For Cartesian} \quad (11)$$

$$|u_s^W\rangle = \sum_{\mathbf{R}} e^{-i\mathbf{k} \cdot (\mathbf{r} - \mathbf{R} - \tau_s)} |\mathbf{R}s\rangle \quad [\text{wft}] \quad (1)$$

τ assume τ_s indep of β ! (Impose it.)

$$A_{\beta, st}(\underline{k}) = \langle u_s^W | i \partial_\beta | u_t^W \rangle$$

$$A_{\beta, st}(\underline{k}) = \frac{1}{N} \sum_{\mathbf{R}', \mathbf{R}''} e^{i\underline{k} \cdot (\underline{r} - \mathbf{R}' - \tau_s)} e^{-i\underline{k} \cdot (\underline{r} - \mathbf{R}'' - \tau_t)} \langle \mathbf{R}'s | i \partial_\beta | \mathbf{R}''t \rangle$$

$$A_{\beta, st}(\underline{k}) = \sum_{\mathbf{R}} e^{i\underline{k} \cdot \mathbf{b}_{st}(\mathbf{R})} \langle 0s | i \partial_\beta | \mathbf{R}t \rangle \quad \downarrow \mathbf{R} = \mathbf{R}'' - \mathbf{R}'$$

$$\text{Let } \underline{\Lambda}_{st} = \langle 0s | \partial_\beta | \underline{\mathbf{R}}t \rangle$$

$$A_{\beta, st}(\underline{k}) = i \sum_{\mathbf{R}} e^{i\underline{k} \cdot \mathbf{b}_{st}(\mathbf{R})} \underline{\Lambda}_{st}(\mathbf{R})$$

$$\begin{aligned} \text{Note } \underline{\Lambda}_{ts}^*(\mathbf{R}) &= \langle \partial_\beta \mathbf{R}s | 0t \rangle \\ &= -\langle \mathbf{R}s | \partial_\beta 0t \rangle \\ &= -\langle 0s | \partial_\beta \mathbf{R}t \rangle \\ &= -\underline{\Lambda}_{st}(-\mathbf{R}) \end{aligned}$$

$$\downarrow \partial_\beta \langle \mathbf{R}s | 0t \rangle = 0$$

$$\Rightarrow \underline{\Lambda}_\beta^+(\mathbf{R}) = -\underline{\Lambda}_\beta(-\mathbf{R})$$

$\therefore A_\beta(\underline{k})$ is Hermitian.

$$\partial_\mu A_{\beta, st}(\underline{k}) = - \sum_{\mathbf{R}} b_{\mu, st}(\mathbf{R}) e^{i\underline{k} \cdot \mathbf{b}_{st}(\mathbf{R})} \underline{\Lambda}_{st}(\mathbf{R})$$

$$\partial_\beta A_{\mu, st}(\underline{k}) = \sum_{\mathbf{R}} e^{i\underline{k} \cdot \mathbf{b}_{st}(\mathbf{R})} \partial_\beta X_{\mu, st}(\mathbf{R})$$

$$\Omega_{\beta\mu, st} = \sum_{\mathbf{R}} e^{i\underline{k} \cdot \mathbf{b}_{st}(\mathbf{R})} \left[\partial_\beta X_{\mu, st}(\mathbf{R}) + b_{\mu, st}(\mathbf{R}) \underline{\Lambda}_{st}(\mathbf{R}) \right]$$

Non-Parag-Lover Ω